# LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

## Quick Revision: Is Matter Around Us Pure?

### Key terms

| Term | Meaning / example |
|---|---|
| Pure substance | Made of only one kind of particle: an element or a compound |
| Homogeneous mixture | Uniform composition throughout, e.g. salt solution, air |
| Heterogeneous mixture | Non-uniform composition, e.g. sand in water, oil and water |
| Solution | Homogeneous mixture of a solute in a solvent; particles too small to see; no Tyndall effect |
| Suspension | Heterogeneous; particles visible and settle down, e.g. chalk in water |
| Colloid | Particles too small to see but they scatter light (Tyndall effect) and do not settle, e.g. milk, fog |

**Concentration (mass by mass %) = mass of solute ÷ mass of solution × 100**
Mass of solution = mass of solute + mass of solvent

### Separation methods

| Mixture | Method |
|---|---|
| Coloured dye from blue/black ink | Evaporation |
| Cream from milk | Centrifugation |
| Oil and water | Separating funnel |
| Ammonium chloride from common salt | Sublimation |
| Dyes in black ink | Chromatography |
| Acetone and water (boiling points far apart) | Distillation |
| Gases of the air; petroleum fractions (boiling points close) | Fractional distillation |
| Pure copper sulphate from impure sample | Crystallisation |

### Mixture or compound?

- Iron filings mixed with sulphur powder: a **mixture**. A magnet pulls out the iron, and each keeps its own properties.
- Heated together they form a **compound**, iron sulphide: Fe + S → FeS. It has new properties and a fixed composition, and a magnet cannot separate it.

### Quick check

1. A solution contains 40 g of common salt in 320 g of water. Find its concentration.

2. How will you separate a mixture of salt and ammonium chloride?